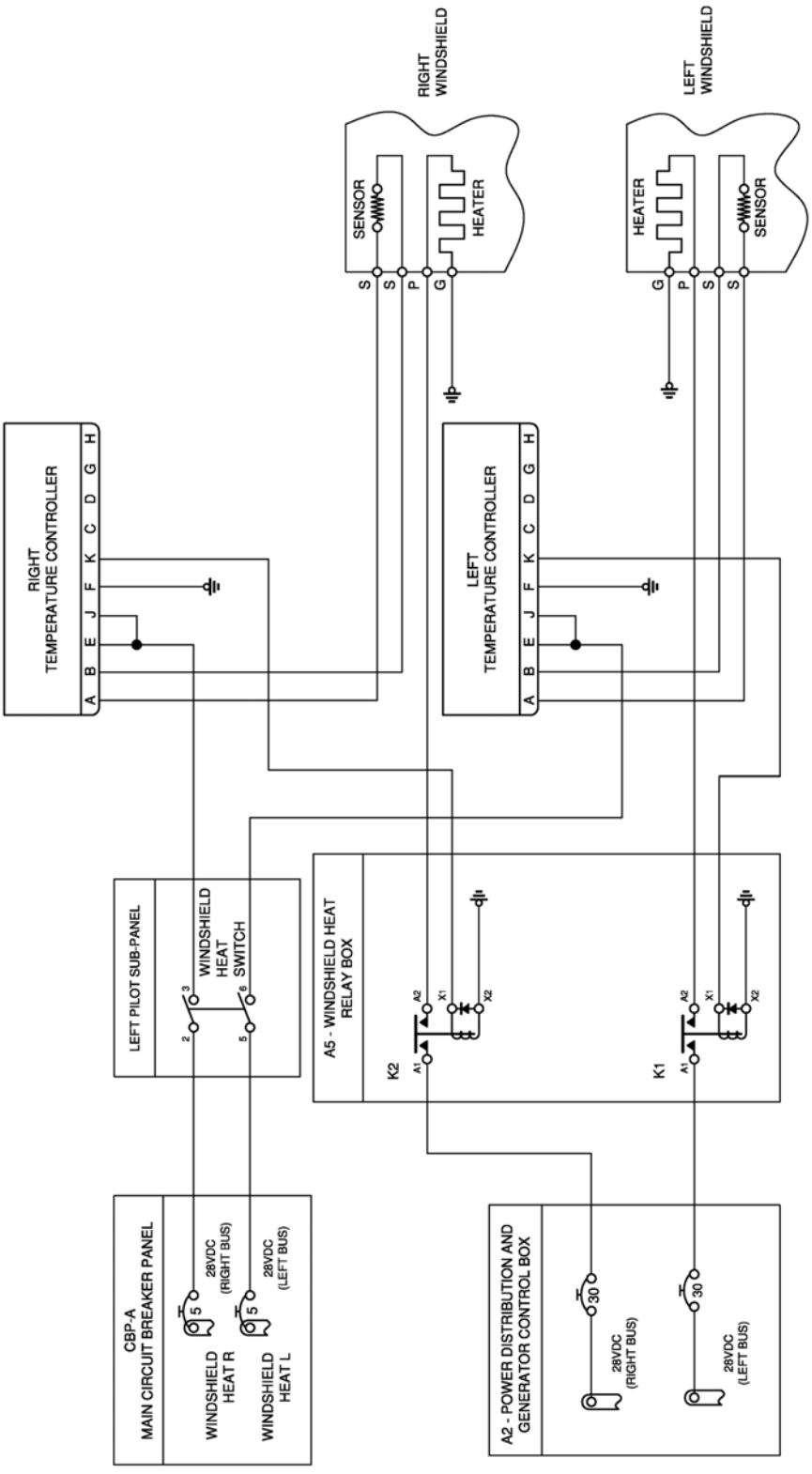
Windshield Heating System (Mod S.O.O. 6187 and Mod 6/2045) – General Data

1. General

The electrically heated windshield system in effect consists of two systems, one for each windshield, both of which are controlled by a single two-pole switch, labelled OFF and ON, located on the W/S HEAT section of the left hand pilot's sub-panel. Each system comprises of a relay, a temperature controller, two circuit breakers, and a windshield manufactured from laminations of glass and vinyl. A fine wire heating element and a sensing element are incorporated in the laminations during the windshield manufacture. The temperature controllers and the relays are installed in the cabin roof under the upholstery. When the windshield heating switch is selected to ON, electrical power from the 28 Volt DC bus bar is routed through the WINDSHIELD HEAT L and R 5-ampere circuit breakers (located on the circuit breaker panel), and the switch, to the left and right temperature controllers. Each temperature controller in conjunction with its related windshield sensing element, provides an output to energize its associated relay coil when the windshield temperature is low. With the relay energized, a circuit from the left (or right) 28 Volt DC bus bar is completed to its associated windshield heating element, through the CB2 (or CB1) 30-ampere circuit breaker located on the power distribution and generator control box. When the windshield reaches its maximum temperature, or when the heating switch is selected OFF, the temperature controller output ceases, the relay (or relays) are deenergized and power is disconnected from the windshield heating element.



Windshield Heating – Electrical Schematic
Figure 1

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Note For the removal and installation of electrically heated windshields refer to 56-10-00, Flight Compartment Windshields – General Data.

2. Adjustment/Test

A. Function Test Windshield Heating System

1) Disconnect electrical wiring at each windshield.

2) Check for continuity between terminals P and G (heating elements) and between terminals S (sensing elements).

Note Resistance between terminals P and G is 1.0 to 1.3 ohms.

3) Reconnect electrical wiring at each windshield.

4) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.). Engage circuit breaker CB2 in power distribution box and circuit breaker WINDSHIELD HEAT L in overhead console.

5) Connect voltmeter across terminals P and G (G is ground) of pilot's windshield.

6) Select windshield heat switch to ON. Check that voltage exists at terminals until temperature of windshield rises and sensing control de-energizes the power relay.

7) When temperature of windshield drops sufficiently to cause the sensing control to reenergize the power relay, check that power is restored to the windshield.

8) Disengage circuit breakers engaged in Para 2.A.4), and engage circuit breaker CB1 in power distribution box and circuit breaker WINDSHIELD HEAT R in left hand pilot's sub-panel. Repeat Para 2.A.5) through Para 2.A.7) for co-pilot's windshield.

9) Return windshield heat switch to OFF and disengage circuit breakers engaged in Para 2.A.8).

10) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

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